Chapter 3: Expectation of Random Variables

Probability and Statistics for Science and Engineering with Examples in R

Hongshik Ahn

(3) Covariance and Independence

Covariance of X_1 and X_2:

$$\mathrm{Cov}\,(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)]$$

If X_1 & X_2 are independent, then

$$\begin{aligned}
&\mathrm{Cov}\,(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)] \\
&= E(X_1X_2 - \mu_1X_2 - \mu_2X_1 + \mu_1\mu_2) \\
&= E(X_1X_2) - \mu_1E(X_2) - \mu_2E(X_1) + \mu_1\mu_2 \\
&= E(X_1)E(X_2) - \mu_1E(X_2) - \mu_2E(X_1) + \mu_1\mu_2 \\
&= \mu_1\mu_2 - \mu_1\mu_2 - \mu_2\mu_1 + \mu_1\mu_2 \\
&= 0
\end{aligned}$$

Alternative form of the covariance:

$$\mathrm{Cov}(X_1, X_2) = E(X_1X_2) - \mu_1\mu_2$$

Theorem 3.2

X and Y are independent $\underset{\not\Leftarrow}{\Rightarrow}$ $\mathrm{Cov}(X,Y) = 0$

<u>Proof</u> $\Rightarrow$) Obvious

$\Leftarrow$) Define

$$f(x,y) = \begin{cases} \frac{1}{4}, & (x,y) = (-4,1),(4,-1),(2,2),(-2,-2) \\ 0, & \text{otherwise} \end{cases}$$

Then

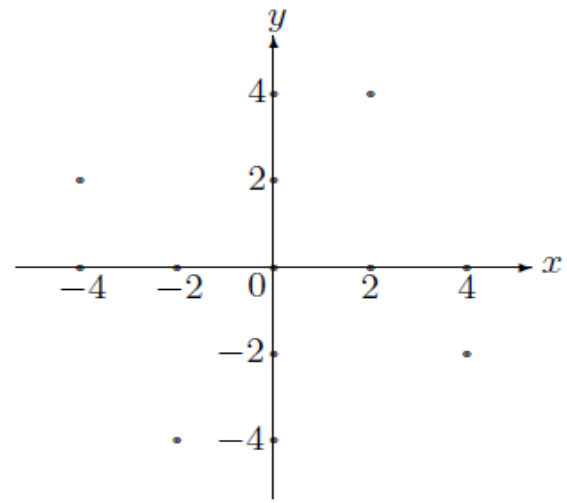
$$E(XY) = \sum_x \sum_y xyf(x,y) = (-4-4+4+4)\cdot\frac{1}{4} = 0$$

$$\mu_X = \sum_x \sum_y xf(x,y) = (-4+4+2-2)\cdot\frac{1}{4} = 0$$

$$\mu_Y = \sum_x \sum_y yf(x,y) = (1-1+2-2)\cdot\frac{1}{4} = 0$$

$$\mathrm{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = 0$$

Theorem 3.2 (continued)



X and Y are independent $\begin{matrix}\Rightarrow \\ \not\Leftarrow\end{matrix}$

$$\mathrm{Cov}(X,Y) = 0$$

<u>Proof</u> $\Leftarrow$)

$$f(x,y) = \begin{cases} \frac{1}{4}, & (x,y) = (-4,1),(4,-1),(2,2),(-2,-2) \\ 0, & \text{otherwise} \end{cases}$$

$$\mathrm{Cov}(X,Y) = E(XY) - \mu_X\mu_Y = 0$$

$$f_X(-4) = f_Y(1) = \frac{1}{4}, f(-4,1) = \frac{1}{4}$$

$$f(-4,1) = \frac{1}{4} \neq f_X(-4)f_Y(1) = \frac{1}{4}\cdot\frac{1}{4} = \frac{1}{16}$$

$\therefore$ X and Y are not independent .

- If X and Y are normal r.v.'s, then the iff condition is satisfied.

<u>Theorem 7.4.4</u>

(1) $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$

<u>Proof</u>

$\mathrm{Cov}(X, Y) = E(XY) - \mu_X \mu_Y = E(YX) - \mu_Y \mu_X$
$= \mathrm{Cov}(Y, X)$

(2) $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$

<u>Proof</u>

$\mathrm{Cov}(X, X) = E(X^2) - {\mu_X}^2 = \mathrm{Var}(X)$

Theorem 3.3 (continued)

(3) $\mathrm{Cov}(aX + bY, cX + dY)$

$$= ac\mathrm{Var}(X) + bd\mathrm{Var}(Y) + (ad + bc)\mathrm{Cov}(X, Y)$$

Proof

$\mathrm{Cov}(aX + bY, cX + dY)$

$= E[(aX + bY)(cX + dY)]$

$- [E(aX + bY)][E(cX + dY)]$

$= [acE(X^2) + (ad + bc)E(XY) + bdE(Y^2)]$

$- [ac\{E(X)\}^2 + (ad + bc)E(X)E(Y) + bd\{E(Y)\}^2]$

$= ac[E(X^2) - \{E(X)\}^2] + bd[E(Y^2) - \{E(Y)\}^2]$

$+ (ad + bc)[E(XY) - E(X)E(Y)]$

$= ac\mathrm{Var}(X) + bd\mathrm{Var}(Y) + (ad + bc)\mathrm{Cov}(X, Y)$

Theorem 3.3 (continued)

(4) If X_i is a r.v. and a_i is constant, $i = 1, \cdots, n$, then

$$\mathrm{Var}\left(\sum_{i=1}^{n} a_i X_i\right) = \sum_{i=1}^{n} a_i^2 \mathrm{Var}(X_i) + 2\sum_{i<j}\sum a_i a_j \mathrm{Cov}\left(X_i, X_j\right)$$

Correlation Coefficient

Correlation coefficient:

$$\rho = \frac{\mathrm{Cov}(X_1, X_2)}{\sigma_{X_1}\sigma_{X_2}}$$

$$-1 \leq \rho \leq 1$$

<u>Proof</u>

$$0 \leq \mathrm{Var}\left(\frac{X}{\sigma_X} + \frac{Y}{\sigma_Y}\right) = \frac{\mathrm{Var}(X)}{{\sigma_X}^2} + \frac{\mathrm{Var}(Y)}{{\sigma_Y}^2} + 2\frac{\mathrm{C}ov(X,Y)}{\sigma_X\sigma_Y}$$

$$= 2(1+\rho)$$

$$\therefore \rho \geq -1$$

$$0 \leq \mathrm{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = \frac{\mathrm{Var}(X)}{{\sigma_X}^2} + \frac{\mathrm{Var}(Y)}{{\sigma_Y}^2} - 2\frac{\mathrm{C}ov(X,Y)}{\sigma_X\sigma_Y}$$

$$= 2(1-\rho)$$

$$\therefore \rho \leq 1$$

<u>Example 5.2:</u> Find the marginal distribution of X and Y, and $P\left(\frac{1}{2} \leq Y \leq \frac{3}{4}\right)$.

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), & 0 \leq x \leq 2, 0 \leq y \leq 1. \\ 0, & \text{otherwise} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy = \int_0^1 \frac{3}{8}(x+y^2)dy = \frac{3}{8}\left[xy + \frac{y^3}{3}\right]_{y=0}^{1} = \frac{3}{8}\left(x + \frac{1}{3}\right)$$

$$= \begin{cases} \frac{1}{8}(3x+1), & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx = \int_0^2 \frac{3}{8}(x+y^2)dx = \frac{3}{8}\left[\frac{x^2}{2} + xy^2\right]_{x=0}^{2} = \frac{3}{8}(2+2y^2)$$

$$= \begin{cases} \frac{3}{4}(y^2+1), & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

$$P\left(\frac{1}{2} \leq Y \leq \frac{3}{4}\right) = \int_{1/2}^{3/4} f_2(y)dy = \frac{3}{4}\int_{1/2}^{3/4}(y^2+1)dy = \left[\frac{y^3}{4} + \frac{3y}{4}\right]_{y=\frac{1}{2}}^{\frac{3}{4}} = \left(\frac{27}{256} + \frac{9}{16}\right) - \left(\frac{1}{32} + \frac{3}{8}\right)$$

$$= \frac{67}{256}$$

<u>Example 3.14</u> (continued): Find the correlation coefficient of X and Y.

$$f_X(x) = \begin{cases} \frac{1}{8}(3x+1), & 0 \le x \le 2 \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx = \int_0^2 \frac{1}{8}(3x^2 + x)\, dx = \frac{1}{8}\left[x^3 + \frac{x^2}{2}\right]_0^2 = \frac{1}{8}(8+2) = \frac{5}{4}$$

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \int_0^2 \frac{1}{8}(3x^3 + x^2)\, dx = \frac{1}{8}\left[\frac{3x^4}{4} + \frac{x^3}{3}\right]_0^2 = \frac{1}{8}\left(12 + \frac{8}{3}\right) = \frac{11}{6}$$

$$Var(X) = E(X^2) - [E(X)]^2 = \frac{11}{6} - \left(\frac{5}{4}\right)^2 = \frac{13}{48}$$

Example 3.14 (continued)

$$f_Y(y) = \begin{cases} \frac{3}{4}(y^2+1), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$

$$E(Y) = \int_{-\infty}^{\infty} y f_2(y) dy = \int_0^1 \frac{3}{4}(y^3+y)\, dy = \frac{3}{4}\left[\frac{y^4}{4}+\frac{y^2}{2}\right]_0^1 = \frac{3}{4}\left(\frac{1}{4}+\frac{1}{2}\right) = \frac{9}{16}$$

$$E(Y^2) = \int_{-\infty}^{\infty} y^2 f_2(y) dy = \int_0^1 \frac{3}{4}(y^4+y^2)\, dx = \frac{3}{4}\left[\frac{y^5}{5}+\frac{y^3}{3}\right]_0^1 = \frac{3}{4}\left(\frac{1}{5}+\frac{1}{3}\right) = \frac{2}{5}$$

$$Var(Y) = E(Y^2) - [E(X)]^2 = \frac{2}{5} - \left(\frac{9}{16}\right)^2 = \frac{107}{1280}$$

Example 3.14 (continued)

$$f(x,y) = \begin{cases} \frac{3}{8}(x+y^2), & 0 \le x \le 2, 0 \le y \le 1. \\ 0, & \text{otherwise} \end{cases}$$

$$\mathrm{E}(XY) = \int_{-\infty}^{\infty} xyf(x,y)dxdy = \int_0^1 \int_0^2 \frac{3}{8}(x^2y + xy^3)\,dxdy = \int_0^1 \left[\frac{3}{8}\left(\frac{x^3y}{3} + \frac{x^2y^3}{2}\right)\Big|_0^2\right] dy$$

$$= \int_0^1 \left(y + \frac{3y^3}{4}\right) dy = \left(\frac{y^2}{2} + \frac{3y^4}{16}\right)\Bigg|_0^1 = \frac{1}{2} + \frac{3}{16} = \frac{11}{16}$$

$$Cov(X,Y) = E(XY) - E(X)E(Y) = \frac{11}{16} - \left(\frac{5}{4}\right)\left(\frac{9}{16}\right) = -\frac{1}{64}$$

$$\rho = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{-\frac{1}{64}}{\sqrt{\left(\frac{13}{48}\right)\left(\frac{107}{1280}\right)}} = -0.104$$

- Let $X_1, \cdots, X_n$ be n r.v.s & $a_1, \cdots, a_n$ be constants. Then

$$Y = a_1 X_1 + \cdots + a_n X_n = \sum_{i=1}^{n} a_i X_i$$

is called a <u>linear combination</u> of the X_i's.

eg) If $a_1 = \cdots = a_n = 1$, then $Y = \sum_{i=1}^{n} X_i$

If $a_1 = \cdots = a_n = \frac{1}{n}$, then $Y = \bar{X}$

- Let X_i have mean μ_i & variance σ_i^2 for $i = 1, 2, \cdots, n$. Then
  - $E(a_1 X_1 + \cdots + a_n X_n) = a_1 E(X_1) + \cdots + a_n E(X_n)$
    i.e., $\mu_Y = \sum_{i=1}^{n} a_i \mu_i$
  - If $X_1, \cdots, X_n$ are independent,
    $\mathrm{Var}(a_1 X_1 + \cdots + a_n X_n) = a_1^2 Var(X_1) + \cdots + a_n^2 Var(X_n)$
    i.e., $\sigma_Y^2 = \sum_{i=1}^{n} a_i^2 \sigma_i^2$

Example 3.15

Prices of 3 different brands of toothpastes: \$2.40, \$2.70, \$2.90 per pack

X_i: total amount of brand i purchased on a particular day, $i = 1, 2, 3$

$X_i's$ are independent r.v.'s with $\mu_1 = 500, \mu_2 = 250, \mu_3 = 200$;

$$\sigma_1 = 50, \quad \sigma_2 = 40, \quad \sigma_3 = 25$$

The revenue from sales: $Y = 2.4X_1 + 2.7X_2 + 2.9X_3$

What are the mean and standard deviation of Y?

$$E(Y) = 2.4\mu_1 + 2.7\mu_2 + 2.9\mu_3$$
$$= 2.4(500) + 2.7(250) + 2.9(200) = \$2{,}455$$
$$Var(Y) = (2.4)^2{\sigma_1}^2 + (2.7)^2{\sigma_2}^2 + (2.9)^2{\sigma_3}^2$$
$$= (2.4)^2(2500) + (2.7)^2(1600) + (2.9)^2(625)$$
$$= \$31{,}320.25$$
$$\sigma_Y = \sqrt{31{,}320.25} = \$176.98$$

Special case: $n = 2$, $a_1 = 1$ & $a_2 = -1$. Then

$$\mathrm{Y} = a_1 X_1 + a_2 X_2 = X_1 - X_2$$

$$E(X_1 - X_2) = \mu_1 - \mu_2$$

If X_1 & X_2 are independent, then $\mathrm{Var}(X_1 - X_2) = \sigma_1^2 + \sigma_2^2$.

<u>Example 3.16</u> The tar contents of the cigarettes from two different brands are known to be different. Let X_1 and X_2 be the tar contents of the cigarettes from Brand 1 and Brand 2, respectively. X_1 and X_2 are independent. Let $\mu_1 = 15$ mg, $\mu_2 = 17$ mg, $\sigma_1 = 1.1$ mg and $\sigma_2 = 1.4$ mg.

$$\Rightarrow E(X_1 - X_2) = \mu_1 - \mu_2 = 15 - 17 = -2 \text{ mg}$$

$$Var\,(X_1 - X_2) = \sigma_1^2 + \sigma_2^2 = (1.1)^2 + (1.4)^2 = 3.17 \text{ mg}$$

$$\sigma_{X_1 - X_2} = \sqrt{3.17} = 1.78 \text{ mg}$$

If $X_1, \cdots, X_n$ are independent and $X_i \sim N(\mu_i, \sigma_i^2)$ for $i = 1,2, \cdots, n$,

$$Y = a_1 X_1 + \cdots + a_n X_n \sim N(a_1\mu_1 + \cdots + a_n\mu_n, a_1^2\sigma_1^2 + \cdots + a_n^2\sigma_n^2)$$

$$X_1 - X_2 \sim N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

<u>Example 3.17</u>

From <u>Example 3.15</u>, suppose X_1, X_2 & X_3 are normal. Then

$$Y = 2.4X_1 + 2.7X_2 + 2.9X_3$$

$$Y \sim N(2.4\mu_1 + 2.7\mu_2 + 2.9\mu_3, 2.4^2\sigma_1^2 + 2.7^2\sigma_2^2 + 2.9^2\sigma_3^2)$$

$$= N(2455, 31320.25)$$

Example 3.18

Let $X_1, \cdots, X_4$ be independent normal random variables and X_i distributed as $N\left(\mu_i, \sigma_i^2\right)$ for $i = 1, \cdots, 4$.

(a) Find $P(\bar{X} < 11)$ when $\mu_1 = \cdots = \mu_4 = 10$ and $\sigma_1^2 = \cdots = \sigma_4^2 = 2$.

(b) Find $P(4X_1 + X_2 - X_3 - 4X_4 < 3)$ when $\mu_1 = \cdots = \mu_4 = 10$ and $\sigma_1^2 = \cdots = \sigma_4^2 = 2$.

(c) Find $P(35 < X_1 + \cdots + X_4 < 45)$ when $\mu_1 = \mu_2 = 8$, $\mu_3 = \mu_4 = 12$, $\sigma_1^2 = \sigma_2^2 = 1$ and $\sigma_3^2 = \sigma_4^2 = 2$.

Answer to Example 3.18

(a) $\bar{X} \sim N(10, 2/4) = N(10, 0.5)$

$$P(\bar{X} < 11) = P\left(Z < \frac{11-10}{\sqrt{0.5}}\right) = \Phi(1.414) = 0.9213$$

(b) $4X_1 + X_2 - X_3 - 4X_4 \sim N(4\mu_1 + \mu_2 - \mu_3 - 4\mu_4, 16\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + 16\sigma_4^2)$

$$= N(0, 68)$$

$$P(4X_1 + X_2 - X_3 - 4X_4 < 3) = P\left(Z < \frac{3}{\sqrt{68}}\right) = \Phi(0.364) = 0.642$$

(c) $X_1 + \cdots + X_4 \sim N(\mu_1 + \cdots + \mu_4, \sigma_1^2 + \cdots + \sigma_4^2) = N(2 \cdot 8 + 2 \cdot 12, 2 \cdot 1 + 2 \cdot 2) = N(40, 6)$

$$P(35 < X_1 + \cdots + X_4 < 45) = P\left(\frac{35-40}{\sqrt{6}} < Z < \frac{45-40}{\sqrt{6}}\right)$$

$$= P(-2.04 < Z < 2.04) = \Phi(2.04) - \Phi(-2.04)$$

$$= \Phi(2.04) - [1 - \Phi(2.04)] = 2\Phi(2.04) - 1 = 2(0.9793) - 1 = 0.9586$$

Example 3.19

(a) U and V: be independent N(0,1) r.v.'s.

Set $Z = \rho U + \sqrt{1-\rho^2}V, -1 < \rho < 1.$

$$E(Z) = \rho E(U) + \sqrt{1-\rho^2}E(V) = 0$$

$$\mathrm{Var}(Z) = \rho^2 \mathrm{Var}(U) + (1-\rho^2)\mathrm{Var}(V) = 1$$

$$\therefore Z \sim \mathrm{N}(0,1)$$

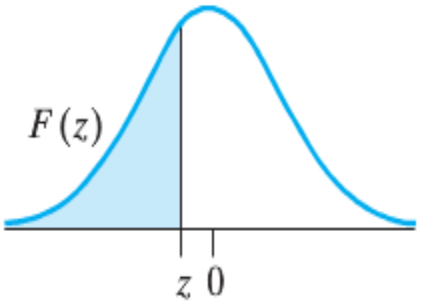
(b) $\mathrm{Cov}(U,Z) = \mathrm{Cov}\left(U, \rho U + \sqrt{1-\rho^2}V\right)$

$$= \rho \mathrm{Var}\,(U) + \sqrt{1-\rho^2}\,\mathrm{Cov}(U,V)$$

$$= \rho \cdot 1 + \sqrt{1-\rho^2} \cdot 0 = \rho$$

$$\mathrm{Cor}(U,Z) = \frac{\mathrm{Cov}(U,Z)}{\sigma_U \sigma_Z} = \frac{\rho}{1 \cdot 1} = \rho$$

Table A.1 Standard Normal Distribution Table



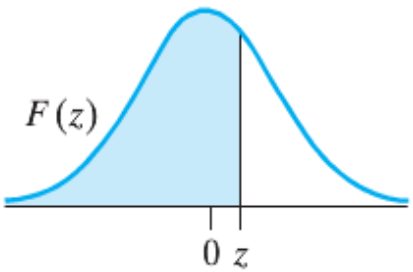
(Area under z curve to the left of z)
$= P(Z < z) = P(Z \leq z)$
$= \Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

z	Area
−3.50	0.00023263
−4.00	0.00003167
−4.50	0.00000340
−5.00	0.00000029

Source: Computed by M. Longnecker using Splus.

Table A.1 Standard Normal Distribution Table



(Area under z curve to the left of z)
$= P(Z < z) = P(Z \leq z)$
$= \Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

z	Area
3.50	0.99976737
4.00	0.99996833
4.50	0.99999660
5.00	0.99999971